

PREETINDERJEET SINGH

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SUMMARY

AI Engineer experienced in building retrieval-augmented and multi-agent systems — agent memory design, tool orchestration, and production deployment — with self-directed study in agent protocols (MCP, ADK).

EDUCATION

Guru Nanak Dev Engineering College (GNDEC), Ludhiana

Aug 2023 – Jun 2027

B.Tech in Computer Science: 8 CGPA (Till 6th Sem)

SKILLS

Languages: Python, JavaScript

Agent Frameworks & Protocols: LangChain, RAG Pipelines, Multi-Agent Orchestration, MCP & Google ADK (self-study), Prompt Engineering

Memory & Retrieval: Pinecone, Qdrant, Neo4j (Knowledge Graphs), SQLite, Redis, Vector Databases, Embeddings, Short/Long-Term & Episodic Memory Design

Backend & Deployment: FastAPI, Flask, Docker, Socket.IO, JSON-RPC APIs, Cloud Deployment (Render, Railway)

EXPERIENCE

Machine Learning R&D Engineer — Lupex Space

Jun 2026 – Present

- Extended and hardened two internal RAG/agent systems — a retrieval-grounded document generator and a multi-agent automation system — shipping new features and resolving production issues (see Projects)
- Lead a team of ML R&D interns, reporting directly to the co-founder, on simulation-based ML pipelines for aerospace/defense applications
- Research missile classification and hypersonic trajectory prediction models

Contractual Developer — 3S Scholars

Aug 2026 – Present

- Scoped a RAG-based admissions/curriculum AI assistant alongside a QR-code attendance system for a K-12 school, reporting directly to the CEO
- Designing the SQL schema, CSV data pipelines, and role-based access control for the deployment-grade attendance system

Data Science Intern / Research Fellow — Punjab AI Excellence (PAI) — Remote

Jun 2025 – Jul 2025

- Fine-tuned an EfficientNet-B0 transfer-learning model for waste classification, reaching 87.08% test accuracy, then deployed it with TensorFlow Lite for real-time edge inference
- Curated and cleaned a 3,000+ image dataset, correcting class imbalance across five categories

LEADERSHIP & INVOLVEMENT

Student Coordinator, Panel Discussions — ICCT-SD 2026 (GNDEC Ludhiana × NIT Delhi)

- Coordinated panel discussion logistics and speaker/participant interactions, and co-prepared end-to-end reports and questionnaires with faculty

PROJECTS

Multi-Agent Marketing System — 3-agent automation with polyglot memory

LangChain · Flask · Socket.IO · SQLite · Neo4j · Redis · JSON-RPC · Docker

- Engineered visibility into a 4-tier memory architecture (SQLite short-term/long-term/episodic, Neo4j semantic) by tagging each agent action with its memory tiers, exposed via a new live API
- Built the live operations dashboard around this data — real-time memory-tier gauges and per-event memory chips — using Socket.IO for live agent activity streaming

[Live demo — multi-agent-marketing-system.onrender.com]

AI Offer Letter Agent — RAG-based document generation system

FastAPI · React · LangChain · Pinecone · Docker · Render

- Built a retrieval-augmented generation pipeline (PDF → chunk → embed → retrieve) grounding offer-letter generation in company policy documents, with deterministic, non-LLM salary computation
- Added a second, independent LLM chain — a summarization agent — producing candidate-facing summaries of generated letters

[Live demo — ai-offerletter-gen.onrender.com]

PUBLICATIONS

Adaptive Multi-UAV Coordination for Efficient Disaster Response Under Uncertainty — ICAAV 2026, Paper 090

Co-author — DRDO-organized conference, presented Aug 2026

- Co-authored and presented a paper on coordination strategies for multiple autonomous agents (UAVs) operating under uncertainty in disaster response
- Extended abstract accepted for the conference's Book of Abstracts; full paper under further review for Springer proceedings

CERTIFICATIONS & TRAINING

Time Series Analysis (HackerEarth × IIT Guwahati) · Python Programmer Certification (Infosys) · Self-Study: RAG, Google ADK, Model Context Protocol (MCP)